

VALOQ TERMINAL — PRODUCTION INFRASTRUCTURE & VENDOR DIRECTORY

Domain: valoq.co.uk | API: api.valoq.co.uk | Architecture: Next.js 14, FastAPI, Supabase, Redis, Gemini 1.5

Platform / Service	Primary Role	Why We Used It	Actions, Modifications & Configs Applied
Porkbun porkbun.com	<i>Domain Registrar & Edge DNS</i>	Domain acquisition with transparent pricing, zero renewal markups, and free WHOIS privacy.	- Registered root domain: valoq.co.uk. - Configured A Record: @ → 76.76.21.21 (Vercel). - Configured CNAME: www → cname.vercel-dns.com. - Configured CNAME: api → valoq-backend.onrender.com.
Vercel vercel.com	<i>Frontend Hosting & Edge CDN</i>	Global serverless hosting optimized for Next.js with automatic TLS termination.	- Connected GitHub repository: ApexQuantum. - Set Root Directory to apex-frontend. - Injected env: NEXT_PUBLIC_SUPABASE_URL & ANON_KEY. - Bound custom domain: valoq.co.uk and configured auto-redirect from www.
Render render.com	<i>Backend Compute & Uvicorn Engine</i>	Container and native Python web service hosting with persistent execution and zero-downtime deploys.	- Created Web Service: valoq-backend pointing to apex-backend. - Command: uvicorn main:app --host 0.0.0.0 --port \$PORT. - Bound custom subdomain: api.valoq.co.uk. - Set runtime env vars: Supabase keys, Redis URL, Gemini Key.
Supabase supabase.com	<i>Managed PostgreSQL & User State</i>	Serverless PostgreSQL with instant REST/GraphQL APIs and Row-Level Security (RLS).	- Provisioned project: mzbcbvazvagesgymqzhdi. - Ran migrations for watchlist (CRUD + custom buy targets). - Ran migrations for equity_prices (OHLCV candles with composite index). - Ran migrations for equity_financials (normalized multi-year filings).
Upstash upstash.com	<i>Serverless Hot-Tier Redis</i>	Low-latency external caching to shield yfinance and Supabase from rate limits and duplicate queries.	- Created Redis database with TLS connection string (rediss://). - Wired REDIS_URL to Render environment. - Programmed dual-cache fallback in backend (Redis → TTLCache in memory) with 60s TTL on quotes and 300s TTL on search.
Google AI Studio aistudio.google.com	<i>Gemini 1.5 Flash Reasoning API</i>	Sub-second LLM streaming for on-demand investment memos, SEC 10-K risk audits, and Bull/Bear models.	- Generated and rotated production API key. - Injected GEMINI_API_KEY into Render backend. - Wired backend route /api/v1/stock/ai-stream with Server-Sent Events (SSE) token generation and frontend live typewriter rendering.
UptimeRobot uptimerobot.com	<i>Cold-Start Preventer & Health Monitor</i>	Free-tier cloud platforms hibernate after 15 minutes of inactivity; automated pings prevent spin-down delays.	- Configured HTTPS monitor targeting https://valoq-backend.onrender.com/. - Scheduled 5-minute automated HTTP GET polling interval. - Activated alert dispatch for backend 500 error drops or DNS timeouts.
GitHub github.com	<i>Source Control & CI/CD Pipeline</i>	Single source of truth repository triggering automated webhooks for staging and production builds.	- Monorepo structure (apex-frontend, apex-backend, nginx). - Wired Git push triggers: pushes to main trigger Vercel Edge builds and Render Docker redeployments automatically.